

Section A: Pure Mathematics [35 marks]

- 1** Find, algebraically, the set of values of m for which the curves $y = x^2 - 2mx - 4.5$ and $y = 2x^2 - 3x + m$ do not intersect. [4]

Solutions*

$$2x^2 - 3x + m = x^2 - 2mx - 4.5$$

$$x^2 + (2m - 3)x + m + 4.5 = 0$$

As the curves do not intersect (there is no real root), $b^2 - 4ac < 0$

$$(2m - 3)^2 - 4(m + 4.5) < 0$$

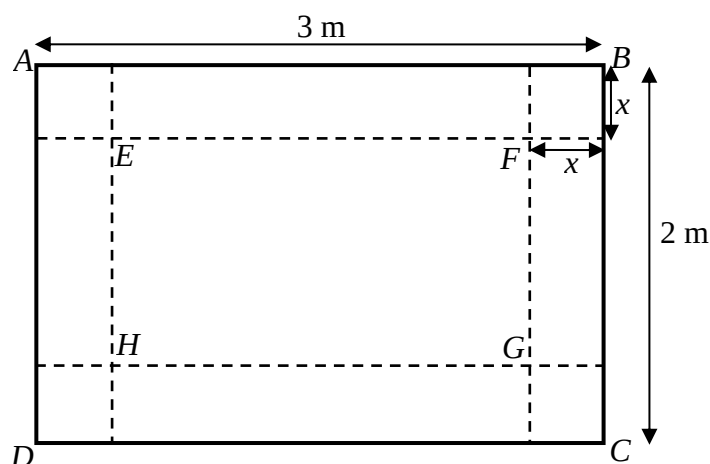
$$4m^2 - 12m + 9 - 4m - 18 < 0$$

$$4m^2 - 16m - 9 < 0$$

$$(2m + 1)(2m - 9) < 0$$

$$-0.5 < m < 4.5$$

2



The diagram shows a rectangular piece of metal sheet of length 3 m and breadth 2 m. A square of side x m is removed from each corner of $ABCD$. The remaining shape is now folded along EF , FG , GH and HE to form an open rectangular box of height x m.

- (i) Show that the volume, $V \text{ m}^3$, of the box is given by $V = 4x^3 - 10x^2 + 6x$. [2]
- (ii) Without using a calculator, find in surd form the values of x that give stationary values of V . Determine the nature of these stationary values. [4]
- (iii) Find the set of values of x for which V is decreasing as x varies. [1]

Solutions*

$$(i) V = x(3 - 2x)(2 - 2x)$$

$$= 4x^3 - 10x^2 + 6x$$

$$(ii) \frac{dV}{dx} = 12x^2 - 20x + 6$$

For stationary points, $\frac{dV}{dx} = 0$

$$12x^2 - 20x + 6 = 0$$

$$6x^2 - 10x + 3 = 0$$

$$x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(6)(3)}}{2(6)} = \frac{10 \pm \sqrt{28}}{12} = \frac{5 \pm \sqrt{7}}{6}$$

Check the nature of stationary value

It is easier to use second derivative test.

$$\text{When } x = \frac{5 + \sqrt{7}}{6} = 1.27 > 1,$$

$$\frac{d^2V}{dx^2} = 24x - 20 = 24\left(\frac{5 + \sqrt{7}}{6}\right) - 20 = 4\sqrt{7} > 0$$

$$V \text{ is minimum when } x = \frac{5 + \sqrt{7}}{6}.$$

Note that in the context of this question, $0 < x < 1$, or V is

negative so $x = \frac{5 + \sqrt{7}}{6}$ is rejected as a stationary value.

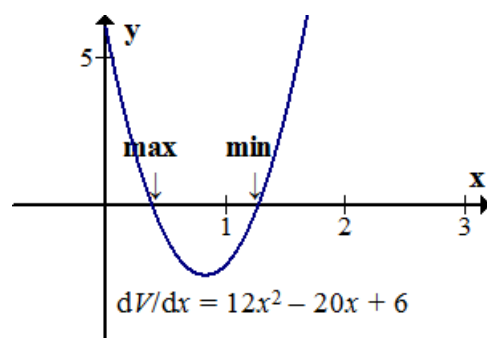
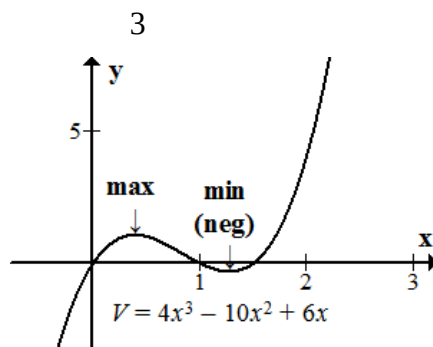
$$\text{When } x = \frac{5 - \sqrt{7}}{6} = 0.39 < 1,$$

$$\frac{d^2V}{dx^2} = 24x - 20 = 24\left(\frac{5 - \sqrt{7}}{6}\right) - 20 = -4\sqrt{7} < 0$$

$$V \text{ is maximum when } x = \frac{5 - \sqrt{7}}{6}$$

$$(iii) V \text{ is decreasing, } \frac{dV}{dx} = 12x^2 - 20x + 6 < 0$$

$$\frac{5 - \sqrt{7}}{6} < x < 1 \text{ (Will accept } < \frac{5 + \sqrt{7}}{6} \text{ because exam condition)}$$

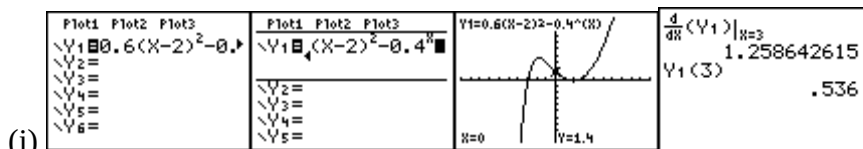


3 The curve C has equation $y = 0.6(x - 2)^2 - 0.4^x$.

(i) Find the equation of the tangent to C at the point P where $x = 3$. Give your answer in the form $y = mx + c$, with m and c correct to 4 decimal places. [3]

(ii) This tangent meets the y -axis at point A and the line $y = 2x$ at point B . If the ratio $PA : AB = 1 : n$, using similar triangles or otherwise, find n to 4 decimal places. [4]

Solutions*



Using GC, at the point P when $x = 3$,

the numerical value of the gradient of C , $\frac{dy}{dx} = 1.258642615$

H2: $\frac{dy}{dx} = 1.2(x - 2) - 0.4^x \ln 0.4$. When $x = 3$, $\frac{dy}{dx} = 1.258642607$ (using calculator)

When $x = 3$, $y = 0.536$

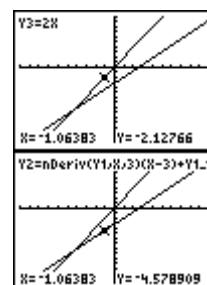
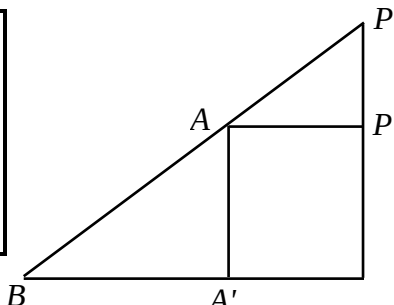
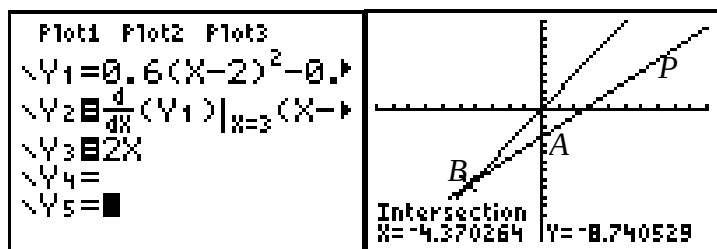
$y - (0.536) = 1.258642615(x - 3)$

$y = 1.258642615x - 3.2399278452$

The equation of the tangent to C at the point P is

$y = 1.2586x - 3.2399$

(ii) Enter the equation of the tangent into GC: $Y2 = \frac{d}{dX}(Y1)|_{X=3}(X-3) + Y1(3)$.



When the tangent meet the y -axis at point A , $x = 0$, $y = -3.2399278452$.

When the tangent intersect with $y = 2x$ at point B , by GC, $x = -4.370264$, $y = -8.740529$

OR Solving $2x = 1.258642615x - 3.2399278452$

$0.741357385x = -3.2399278452$

$x = -4.370264478$

Now by similar triangles, the ratio $PA : AB = P'A : A'B$

$= 3 - 0 : 0 - (-4.370264) = 3 : 4.370264 = 1 : 1.456754667$

$\therefore n = 1.456754667 = 1.4568$ (4dp)

Otherwise find the lengths of PA and AB .

4 (a) Differentiate $\ln\left(\frac{e^{2\sqrt{x}}}{(5x^4-1)^3}\right)$ with respect to x . [3]

(b) Show that $\frac{3}{x-3} - \frac{4}{2x+1} = \frac{2x+15}{(x-3)(2x+1)}$. Hence find $\int \frac{4x+30}{(x-3)(2x+1)} dx$. [3]

(c) Given that $y = e^{1-2x^2}$. Find $\frac{dy}{dx}$. Hence find $\int 8x e^{-2x^2+1} + \frac{1}{\sqrt[3]{x}} dx$. [2]

Solutions*

(a) $\ln\left(\frac{e^{2\sqrt{x}}}{(5x^4-1)^3}\right) = \ln(e^{2\sqrt{x}}) - \ln(5x^4-1)^3 = 2\sqrt{x} \ln(e) - 3\ln(5x^4-1) = 2x^{\frac{1}{2}} - 3\ln(5x^4-1)$

ALT MTD(H1 Syllabus) Use law of log $\frac{d}{dx} \left(\ln \left(\frac{e^{2\sqrt{x}}}{(5x^4-1)^3} \right) \right)$ $= \frac{d}{dx} \left(2x^{\frac{1}{2}} - 3\ln(5x^4-1) \right)$ $= x^{-\frac{1}{2}} - \frac{3(5)(4x^3)}{5x^4-1}$ $= \frac{1}{\sqrt{x}} - \frac{60x^3}{5x^4-1}$	$\frac{d}{dx} \left(\ln \left(\frac{e^{2\sqrt{x}}}{(5x^4-1)^3} \right) \right) = \text{Using quotient rule (Not required in H1)}$ $\frac{1}{\frac{e^{2\sqrt{x}}}{(5x^4-1)^3}} \left(\left(2 \left(\frac{1}{2} x^{-\frac{1}{2}} \right) e^{2\sqrt{x}} \right) (5x^4-1)^3 - e^{2\sqrt{x}} \left(3(5x^4-1)^2 (4(5x^3)) \right) \right)$ $= \frac{e^{2\sqrt{x}} (5x^4-1)^3}{e^{2\sqrt{x}}} \left(\frac{x^{-\frac{1}{2}} (5x^4-1)^3 - (60x^3 (5x^4-1)^2)}{(5x^4-1)^3} \right)$ $= \left(\frac{x^{-\frac{1}{2}} (5x^4-1)^3 - (60x^3 (5x^4-1)^2)}{(5x^4-1)^3} \right)$
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(b) $\frac{3}{x-3} - \frac{4}{2x+1} = \frac{3(2x+1) - 4(x-3)}{(x-3)(2x+1)} = \frac{6x+3-4x+12}{(x-3)(2x+1)} = \frac{2x+15}{(x-3)(2x+1)}$

$\int_2^4 \frac{4x+30}{(x-3)(2x+1)} dx = 2 \int_2^4 \frac{3}{x-3} - \frac{4}{2x+1} dx$

$= 2 \left[3\ln|x-3| - \frac{4}{2} \ln|2x+1| \right] + C = 6\ln|x-3| - 4\ln|2x+1| + C$

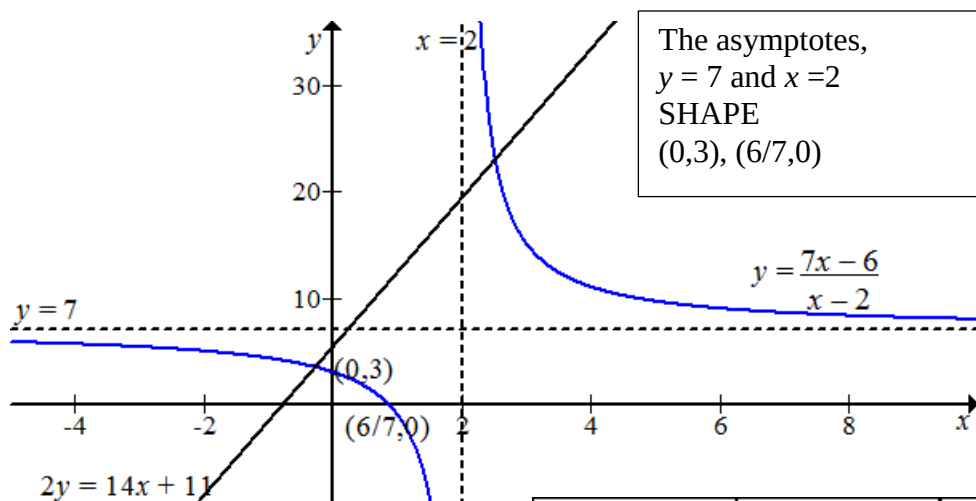
(c)(i) $y = e^{1-2x^2}$ $\frac{dy}{dx} = (-4x)e^{1-2x^2}$ $= -4xe^{1-2x^2}$	(ii) $\int 8x e^{-2x^2+1} + \frac{1}{\sqrt[3]{x}} dx = -2 \int (-4)x e^{-2x^2+1} dx + \int x^{-\frac{1}{3}} dx$ $= -2e^{1-2x^2} + \frac{x^{\frac{2}{3}}}{\frac{2}{3}} + C = -2e^{1-2x^2} + \frac{3x^{\frac{2}{3}}}{2} + C$
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- 5 (i) Sketch the graph of $y = \frac{7x-6}{x-2}$, stating the equations of any asymptotes and the coordinates of any points of intersection with the axes. [3]
- (ii) Find the exact x -coordinates of the points of intersection of $y = \frac{7x-6}{x-2}$ and the line $2y = 14x + 11$. Hence solve the inequality $2\left(\frac{7x-6}{x-2}\right) \leq 14x + 11$. [3]
- (iii) Write down as integral an expression for the area of the region bounded by $y = \frac{7x-6}{x-2}$ and $2y = 14x + 11$ and the line $x = K$ for $K \geq 3$. If the area is 10, find the value of K to 3 decimal places. [3]

Solutions*

$$(i) y = \frac{7x-6}{x-2} = 7 + \frac{8}{x-2}$$

The asymptotes are $y = 7$ and $x = 2$. The coordinates of axial intercepts are $(0,3)$ and $(6/7,0)$



$$(ii) 2\left(\frac{7x-6}{x-2}\right) = 14x + 11$$

$$2(7x-6) = (x-2)(14x+11) \Rightarrow 14x-12 = 14x^2-17x-22$$

$$14x^2-31x-10=0 \quad [\text{equating and obtained quadratic eq}]$$

$(7x+2)(2x-5)=0$	or $x = \frac{31 \pm \sqrt{31^2 - 4(14)(-10)}}{2(14)} = \frac{31 \pm \sqrt{1521}}{28} = \frac{31 \pm 39}{28} = -\frac{8}{28} \text{ or } \frac{70}{28}$
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$$x = -\frac{2}{7} \text{ or } \frac{5}{2} \text{ or } 2.5$$

$$\text{Hence from the graph, } 2\left(\frac{7x-6}{x-2}\right) \leq 14x + 11$$

$$\Rightarrow -\frac{2}{7} \leq x < 2 \text{ or } x \geq 2.5 \quad [\text{Take note of the asymptotes } x = 2]$$

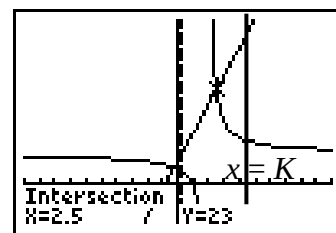
$$\begin{aligned}
 \text{(iii)} \quad \int_{2.5}^K 7x + 5.5 - \left(\frac{7x-6}{x-2} \right) dx &= \int_{2.5}^K 7x + 5.5 - \left(7 + \frac{8}{x-2} \right) dx \\
 &= \int_{2.5}^K 7x - 1.5 - \frac{8}{x-2} dx = \left[x^2 - 1.5x - 8 \ln|x-2| \right]_{2.5}^K \\
 &= K^2 - 1.5K - \ln|K-2| - (2.5^2 - 1.5(2.5) - \ln|2.5-2|)
 \end{aligned}$$

Given that the area is 10,

$$K^2 - 1.5K - \ln|K-2| - (2.5^2 - 1.5(2.5) - \ln|2.5-2|) = 10$$

$$K^2 - 1.5K - \ln|K-2| = 12.5 - \ln 0.5 = 13.19314718$$

By GC, $K = 4.585$ (3 decimal places)



Plot1 Plot2 Plot3 $\sqrt{1} \square x^2 - 1.5x - \ln(x)$ $\sqrt{2} \square 2.5 - \ln(.5)$ $\sqrt{3} =$ $\sqrt{4} =$ $\sqrt{5} =$ $\sqrt{6} =$ Intersection X=4.5847457 Y=13.193147	$ \begin{aligned} &7x + 5.5 - \left(\frac{7x-6}{x-2} \right) \\ &= \frac{(7x+5.5)(x-2) - (7x-6)}{x-2} \\ &= \frac{7x^2 - 14x + 5.5x - 11 - 7x + 6}{x-2} \\ &= \frac{7x^2 - 15.5x - 5}{x-2} = 7x - 1.5 - \frac{8}{x-2} \end{aligned} $	$ \begin{array}{r} 7x - 1.5 \\ x-2 \overline{) 7x^2 - 15.5x - 5} \\ \underline{7x^2 - 14x} \\ -1.5x \\ \underline{-1.5x + 3} \\ -8 \end{array} $
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Section B: Statistics [60 marks]

- 6 At a secondary school Open House, the Principal wishes to find out how many primary school students would like to join her school.
- Explain how a systematic sample could be obtained from 5% of the primary school students. [2]
 - Give an advantage of using systematic sampling in this context. [1]
 - Give a disadvantage of using stratified sampling in this context. [1]

Solutions*

- (i) Assume that there are n primary school students. The sample size is 5% of $n = 0.05n$.

$$\text{The interval} = \frac{n}{0.05n} = 20.$$

The surveyor could interview every $\left(\frac{n}{0.05n} \right)^{\text{th}} = 20^{\text{th}}$ person as the primary school students leave the school, starting from a randomly chosen number from **1st to 20th**.

(ii) An advantage is the students survey are distributed throughout the time of the open day, primary school students are present and asked to fill a simple survey on the spot so the survey less time consuming.

(iii) It is difficult or time-consuming to obtain the information needed in sampling frame for each of the strata. For example, difficult to determine the size of each stratum say by school, gender or race.

It is difficult and more time-consuming to obtain the information from the selected primary school students from the stratification.

OR Any suitable disadvantage

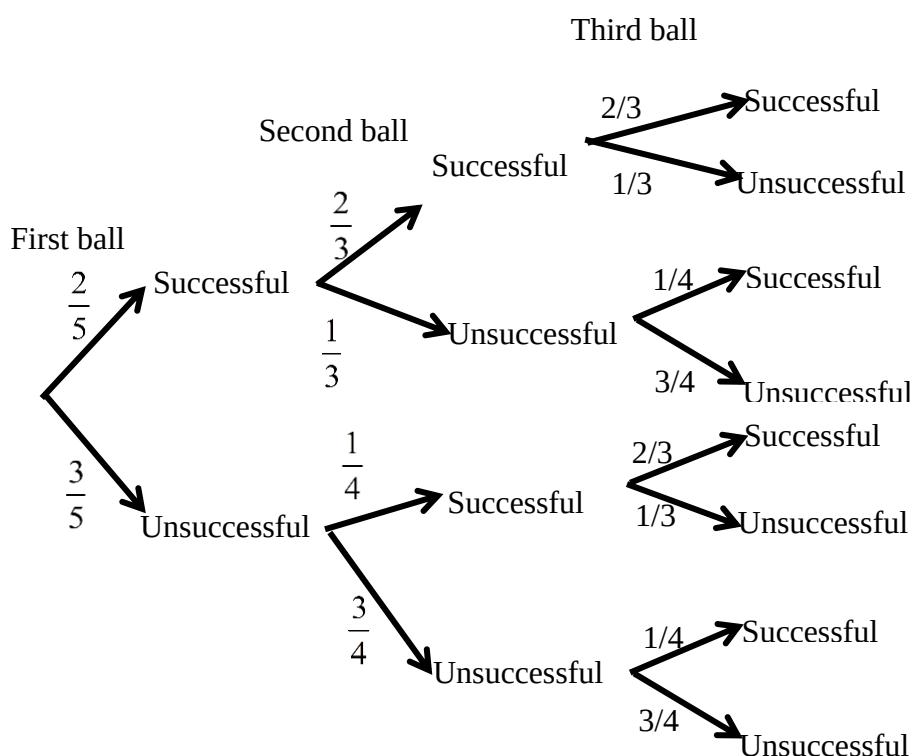
- 7 A player throws three balls at a target. The probability that the player is successful in hitting the target with his first throw is $\frac{2}{5}$. Subsequently, the probability of hitting the target is $\frac{2}{3}$ if the preceding throw is successful and $\frac{1}{4}$ if the preceding throw is unsuccessful.

Find the probability that

- (i) the second throw is successful, [2]
 (ii) two out of three throws are successful, [2]
 (iii) the second throw is successful given that two out of the three throws are successful. [2]

Solutions*

(i).



(i) Let C be the event the second throw is successful

$$P(C) = \left(\frac{2}{5}\right)\left(\frac{2}{3}\right) + \left(\frac{3}{5}\right)\left(\frac{1}{4}\right) = \left(\frac{4}{15}\right) + \left(\frac{3}{20}\right) = \frac{5}{12}$$

(ii) D be the event two out of three throws are successful

$$P(\text{two out of three throws are successful}) = P(D)$$

$$= \left(\frac{2}{5}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{5}\right)\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) + \left(\frac{3}{5}\right)\left(\frac{1}{4}\right)\left(\frac{2}{3}\right)$$

$$= \left(\frac{4}{45}\right) + \left(\frac{1}{30}\right) + \left(\frac{1}{10}\right) = \frac{2}{9}$$

(iii) $P(C | D) = P(C \cap D) \div P(D)$

$$= \left[\left(\frac{2}{5}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{3}{5}\right)\left(\frac{1}{4}\right)\left(\frac{2}{3}\right) \right] \div \frac{2}{9} = \left[\left(\frac{4}{45}\right) + \left(\frac{1}{10}\right) \right] \div \frac{2}{9} = \frac{17}{20}$$

- 8 (a) A group of 200 men and women registered for an advanced cooking course in a culinary school. The school tested the group and found that 10 men and 15 women have an extraordinary sense of taste and smell. A person is randomly selected from the group.

The events E and W are defined as follows.

E : the person selected has extraordinary sense of taste and smell.

W : the person selected is a woman.

Find (i) $P(E)$, [1]

(ii) $P(E \cap W)$, [1]

(iii) the number of women in the group if $P(E \cup W) = 0.47$. [2]

- (b) Events K and L are such that $P(L) = 0.3$, $P(K | L) = 0.8$ and $P(L | K) = 0.6$.

Find $P(K)$. [2]

Are events K and L independent? Justify your answer. [1]

Solutions*

(a) (i) $P(E) = \frac{25}{200} = \frac{1}{8}$ (ii) $P(E \cap W) = \frac{15}{200} = \frac{3}{40}$

Let the number of women be n .	Alternative Method
$P(E \cup W) = P(E) + P(W) - P(E \cap W)$	
$0.47 = \frac{25}{200} + \frac{n}{200} - \frac{15}{200}$	
$\frac{n}{200} = 0.47 - \frac{25}{200} + \frac{15}{200} = 0.42$	
$n = 84$	

	Nbr.Men	Nbr.Women	200
Event E	10	15	25
Event $E \cup W$			200×0.47 $= 94$

E

W

10

15

$n - 15$

$n = 94 - 10 = 84$

- (b) $P(L) = 0.3$, $P(K | L) = 0.8$ and $P(L | K) = 0.6$.

$$P(K | L) = \frac{P(K \cap L)}{P(L)} \Rightarrow P(K \cap L) = 0.3 \times 0.8 = 0.24$$

$$P(L | K) = \frac{P(K \cap L)}{P(K)} \Rightarrow P(K) = 0.24 \div 0.6 = 0.4$$

As $P(K \cap L) = 0.24 \neq P(K)P(L) = 0.4 \times 0.3 = 0.12$ OR

$P(L | K) = 0.6 \neq P(L) = 0.3$ OR $P(K | L) = 0.8 \neq P(K) = 0.4$

Events K and L NOT independent.

State in the context of this question, an assumption needed to model G by a binomial distribution. [1]

(iii) more than 13 mugs that are either blue or red. [2]

Another random sample of 100 mugs is drawn. Using a suitable approximation, find the probability that the sample contains more than 20 and at most 50 green mugs. [3]

Assumption: Choosing a green mug is independent from trial to trial. Probability of a green mug selected in each trial is the same.

- $$= 0.661471483 - 0.003308015 = 0.658163468 = 0.658 \quad [\mathbf{M1A1}]$$

- $$= 1 - 0.520657312 = 0.479342688 = 0.479 \quad [1 - \text{binompdf}(20, \frac{2}{3}, 13)]$$

Check for suitable approximation

$$\approx P(20.5 < H < 50.5) \text{ (continuity correction)}$$

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normalcdf
lower:20.5
upper:50.5
 $\mu$ :100/3
 $\sigma$ : $\sqrt{200/9}$ 
Paste
normalcdf(20.5,50.5,100/3, $\sqrt{200/9}$ )
.9966235755
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- Denoting the mass of a ‘Star’ Chocolate bar by x grams, the results of a random sample of 125 ‘Star’ Chocolate bars, are summarised by

$$\sum(x-300) = -50, \sum(x-300)^2 = 851.$$

- (i) Find the unbiased estimates of the population mean and variance. [2]
- (ii) Test at the 5% significance level whether the claim by the consumer watchdog is valid. [4]
- (iii) Explain, in the context of the question, the meaning of 'at the 5% significance level'. [1]
- (iv) Explain whether it is necessary to assume a normal distribution for the test to be valid. [1]
- (v) The company thinks that the claim made by the consumer watchdog is incorrect. Based on the results in part (ii), explain if the company's claim is justified by testing against the alternative hypothesis of the mean is not 300 at 5% significance level. [2]

Solutions*

(i) Unbiased estimate of the population mean, $\bar{x} = -50/125 + 300 = 299.6$

Unbiased estimate of the population variance, $s^2 = \frac{1}{(125-1)} \left(851 - \frac{(-50)^2}{125} \right) = \frac{831}{124} = 6.701612903$

(ii) Test $H_0: \mu = 300$ against $H_1: \mu < 300$ [B1–consumer watchdog claim]

Perform a lower-tailed test at 5% level of significance

Under H_0 , since $n = 125$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(300, \frac{s^2}{125}\right)$ approximately

Test Statistic: $Z = \frac{\bar{X} - 300}{s/\sqrt{125}} \sim N(0,1)$ OR $\bar{X} \sim N\left(300, \frac{s^2}{125}\right)$

Using GC, as p-value = 0.042036357 < 0.05 [M1 Compare p-value and 5%],

reject H_0 . At 5% significance level, there is sufficient evidence to support the claim that the mean mass of 'Star' Chocolate bars is less than 300 gram.

(iii) It means that there is a probability of 0.05 that the test concludes that the mean mass is less than 300 gram when in fact it is not true.

Or there is a probability of 0.05 that the test will conclude that the claim is less than 300 g given that the mean is 300 g.

Level of significance = P(reject H_0 | H_0 is true)

(iv) No, as $n = 125 (> 50)$ is large, by Central Limit Theorem, $\bar{X}$ will be approximately be normally distributed.

(v) Test $H_0: \mu = 300$ against $H_1: \mu \neq 300$

Perform a two-tailed test at 5% level of significance

Under H_0 , $\bar{X} \sim N\left(300, \frac{s^2}{125}\right)$ approximately by Central Limit Theorem,

Test Statistic: $Z = \frac{\bar{X} - 300}{s/\sqrt{125}} \sim N(0,1)$

The (reviewed) p-value = $2 \times 0.042036357 = 0.084072714 (> 0.05)$ because of two-tailed test.

The company does not reject H_0 and conclude at 5% significance level that there is insufficient evidence that to support that the mean mass is not 300.

- 11** A particular orchard supplies mangoes. The mass of a mango has mean 0.6 kg and standard deviation 0.05 kg.

In normal time, the orchard randomly packs the mangoes into ‘large’ plastic bag and ‘small’ plastic bag for sale. (Assume that the mass of plastic bags are negligible in this question.)

Each ‘large’ plastic bag contains 100 mangoes.

- (i) Explain clearly why the distribution of the mass of a randomly chosen ‘large’ plastic bag of mangoes is approximately normal with mean 60 kg and standard deviation 0.5 kg. [1]
- (ii) Find the probability that a randomly chosen large plastic bag of mangoes has a mass exceeding 61 kg. [1]

Mass of each ‘small’ plastic bag of mangoes has an approximately normal distribution with mean 30 kg and variance 0.125 kg^2 .

- (iii) Find the probability that the mass of five randomly chosen ‘large’ plastic bags of mangoes differs from ten times the mass of a randomly chosen ‘small’ plastic bag of mangoes by at least 1 kg. [3]

Each ‘small’ plastic bags and ‘large’ plastic bags of mangoes are sold at \$18 per kg and \$16 per kg respectively. David buys ten ‘small’ plastic bags of mangoes while Peter buys five ‘large’ plastic bags of mangoes.

- (iv) Find the probability that David pays more than Peter by at least \$650. [3]

During a special offer, 36 mangoes are randomly packed into a ‘special’ plastic bag that cost \$ m per kg. Find the maximum value of m such that the probability of a randomly chosen ‘special’ plastic bag of mangoes costing less than \$320, is more than 0.8. Give your answer to 2 decimal places. [3]

Suggested Solution

Let X be the random variable denoting “the mass of a randomly chosen mango”.

Given that $E(X) = 0.6$, $\text{Var}(X) = 0.05^2$

Let L be the random variable denoting “the mass of a randomly chosen large ‘plastic’ bag of mangoes”.

(i) As $n = 100$ is large, Central Limit Theorem states that, then

$$L = X_1 + \dots + X_{100} \sim N(100 \times 0.6, 100 \times 0.05^2) \text{ approximately (Given)}$$

$$L \sim N(60, 0.25)$$

[B1 – As $n = 100$ is large, apply Central Limit Theorem, and ~~$X_1 + \dots + X_{100}$~~ OR ~~$100 \times 0.6, 100 \times 0.05^2$~~]

$$(ii) P(L > 61) = 0.022750 = 0.0228 \quad \text{GC: normalcdf}(61, E99, 60, 0.5)$$

(iii) Let S be the random variable denoting “the mass of a randomly chosen ‘small’ plastic bag of mangoes”. ~~Removed from question~~ ~~$S = X_1 + \dots + X_{50} \sim N(50 \times 0.6, 50 \times 0.05^2)$~~ approximately,

Given $S \sim N(30, 0.125)$ approximately, then

$$L_1 + L_2 + \dots + L_5 \sim N(5 \times 60, 5 \times 0.25) \text{ ie } N(300, 1.25)$$

$$10S \sim N(10 \times 30, 10^2 \times 0.125) \text{ ie } N(300, 12.5)$$

$$\text{Let } X = L_1 + L_2 + \dots + L_5 - 10S \sim N(0, 13.75)$$

$$P(L_1 + L_2 + \dots + L_5 - 10S > 1) + (10S - L_1 + L_2 + \dots + L_5 > 1)$$

Note difference means “more difference as well as less difference”.

$$= P(X > 1) + P(X < -1)$$

$$= 0.393703 + 0.393703 = 0.787$$

(iv) Let D and P be the amount David and Peter pay respectively. Then

$$D = 18(S_1 + \dots + S_{10}) \sim N(18 \times 10 \times 30, 18^2 \times 10 \times 0.125) \text{ ie } N(5400, 405)$$

$$P = 16(L_1 + \dots + L_5) \sim N(16 \times 5 \times 60, 16^2 \times 5 \times 0.25) \text{ ie } N(4800, 320)$$

$$D - P \sim N(600, 725)$$

$P(\text{David pays more than Peter by at least \$650})$

$$= P(D - P > 650) = 0.031659 = 0.0317$$

Let Y be the random variable “price of a ‘special’ plastic bag of mango”.

Then $Y = \$m(X_1 + \dots + X_{36}) \sim N(k(36)(0.6), m^2(36)(0.05^2))$ i.e

$$Y \sim N(21.6m, (0.3m)^2)$$

$$P(Y < 320) > 0.8$$

Standardising,

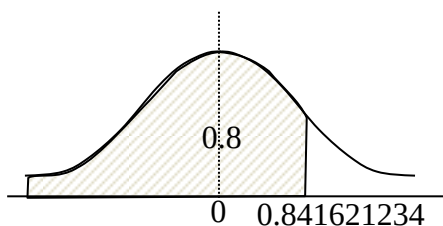
$$P\left(\frac{Y - 21.6m}{0.3m} < \frac{320 - 21.6m}{0.3m}\right) > 0.8$$

$$P\left(Z < \frac{320 - 21.6m}{0.3m}\right) > 0.8$$

$$\text{Now } Z = \frac{Y - 21.6m}{0.3m} \sim N(0,1)$$

$$\frac{320 - 21.6m}{0.3m} > 0.841621234$$

$$[\text{invNorm}(0.8) = 0.841621234]$$



$$320 - 21.6m > 0.25248637k$$

$$320 > 21.85248637m$$

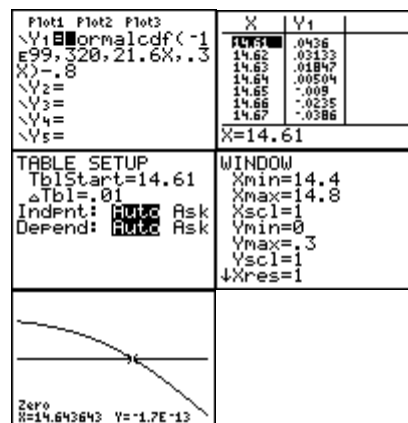
$$14.64364259 > m$$

Using GC,

$$P(Y < 320) > 0.8$$

$$P(Y < 320) - 0.8 > 0$$

$$Y1 = \text{normalcdf}(-E99, 320, 21.6X, .3X) - .8$$



Notice that $14.64 < m < 14.65$, $Y1$ Change from + to -.

$$m = 14.63, P(Y < 320) - 0.8 = 0.01847 > 0$$

$$m = 14.64, P(Y < 320) - 0.8 = 0.00504 > 0$$

$$m = 14.65, P(Y < 320) - 0.8 = -0.009 < 0$$

Hence the greatest $m = 14.64$ (to 2dp)

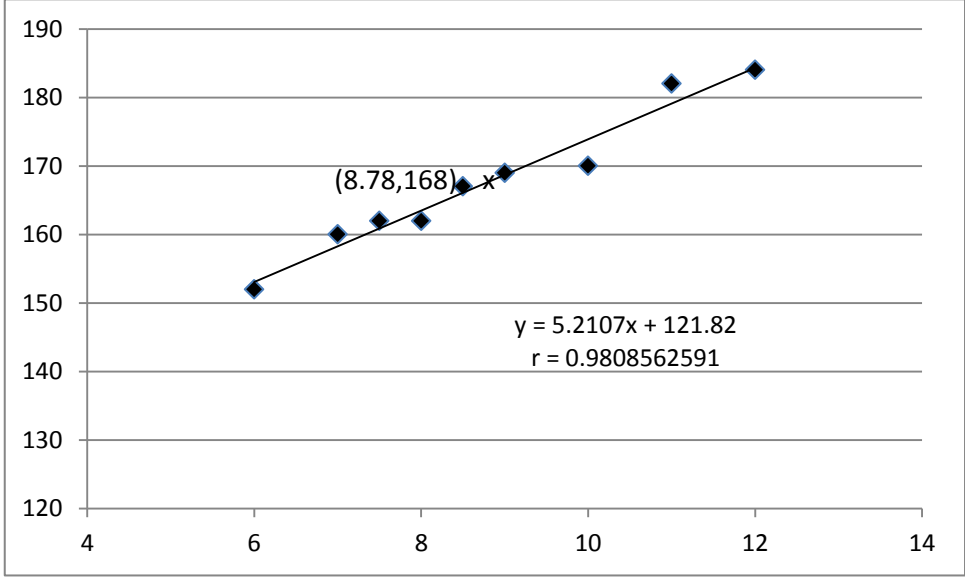
- 12 Shoe size may be used in forensic science to estimate the height of an unknown person.

The table below shows the shoe sizes (inch) and heights (cm) of 9 males.

Shoe Size, x	7.5	7.0	9.0	8.0	8.5	10.0	11.0	12.0	6.0
Height, y	162	160	169	162	167	170	182	184	152

- (i) Draw a scatter diagram for the data. Find $(\bar{x}, \bar{y})$ and mark on your scatter diagram where $(\bar{x}, \bar{y})$ is the mean point of the sample. [2]
- (ii) Find the product moment correlation coefficient and comment on its value in the context of the question. [2]
- (iii) Find the equation of the regression line of y on x , and draw this line on your scatter diagram. [2]
- (iv) Using the regression line from (iii), estimate the shoe size to the nearest half inch when the height of the man is 171 cm. Comment on the reliability of the estimate. [2]
- (v) A new regression line of y on x , $y = 5.07x + 123$ is obtained when another data pair was added. If the shoe size of this data pair is 6.5, find the corresponding height to the nearest cm. [3]
- (vi) Susan found her great-grandfather's shoes in an old rusty iron box and its size is 5.5. She used the regression line in part (v) to predict his height. Comment on the reliability of this prediction. [1]

Solutions*

(i)	 9 PTS increasing trend. $\bar{x} = 8.78$, and $\bar{y} = 168$ ($\bar{x}, \bar{y}$) <u>on the regression line y on x</u> (as well as on <u>x on y</u>) and between $x = 8.5$ and $x = 9.9$	
(ii)	$r = 0.980856259$. There is a strong positive linear correlation between the shoe size and the height. This suggests that the height increases as shoe size increases.	
(iii)	$y = 5.2107x + 121.82$ [from GC] [increasing line passing $(\bar{x}, \bar{y})$ in diagram]	
(iv)	$171 = 5.2107x + 121.82$	

	$x = (171 - 121.82) / 5.2107 = 9.5$ (to nearest half inch) Since $r = 0.981$ indicates there is a strong positive linear correlation between the variables (i.e. the regression lines y on x and x on y are close to each other) and $y = 171$ lies within the data range of x [152cm to 184cm], the estimate is reliable.
(v)	The new regression line is $y = 5.07x + 123$. The new mean point is on this line. New sum of $x = 85.5$ New $\bar{x} = 85.5/10 = 8.55$ $\bar{y} = 5.07\bar{x} + 123 = 5.07(8.55) + 123 = 166.3485$ New sum of $y = 1663.485$ New $y =$ New sum of $y -$ old sum of y $= 1663.485 - 1508 = 155$ (to nearest cm)
(vi)	The estimate is not reliable because shoe size 5.5 cm is outside the data range of the shoe size, so we are extrapolating.